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Data Engineer



Seeking to apply extensive **5+ experience** in managing AWS, Azure, and GCP platforms to architect cloud-based data infrastructures that enhance data availability and scalability.

PROFILE SUMMARY:

- **5+ years** of experience in designing, building, and optimizing data pipelines, focusing on scalable and efficient data infrastructure
- Expertise in **cloud platforms** such as **AWS, Azure, and GCP**, developing and managing cloud-based data solutions to support data analytics and machine learning workloads.
- Proficient in **CI/CD** practices, automating data workflows and deployments using tools like **Jenkins, GitLab**, and **Azure DevOps** to enhance operational efficiency.
- Skilled in **big data technologies** including **Hadoop, Spark**, and **Kafka**, enabling the processing of large-scale data in real-time and batch environments.
- Strong knowledge of **data warehousing** solutions such as **Amazon Redshift, Azure Synapse**, and **Google BigQuery**, optimizing query performance and data storage.
- Hands-on experience in **ETL development** and **data pipeline orchestration** using **Apache Airflow, AWS Glue**, and **Azure Data Factory**, ensuring reliable data ingestion and transformation.
- Adept at ensuring data quality, governance, and security, implementing industry best practices for compliance with **GDPR, HIPAA**, and other data regulations.
- Implement continuous integration and deployment pipelines for GCP infrastructure using **Google Cloud Build** and **Cloud Deployment Manager** to automate infrastructure provisioning and updates. Proficient in **SQL** for database management, data modeling, and complex query optimization across relational databases like **MySQL, PostgreSQL**, and **SQL Server**.
- Hands-on experience with **Amazon EC2, Amazon S3, Amazon RDS, VPC, IAM, Amazon Elastic Load Balancing, Auto Scaling, CloudWatch, SNS, SES, SQS, Lambda, EMR** and other services of the **AWS family**
- Developed web-based applications using **Python, DJANGO, QT, C++, XML, CSS3, HTML5, DHTML, JavaScript** and **jQuery**.
- Proficient with Container systems like **Docker** and Container orchestration like **EC2 Container Service, Kubernetes**, worked with **Terraform**.
- Experience in implementing **Azure data solutions**, provisioning storage account, **Azure Data Factory, SQL server, SQL Databases, SQL Data warehouse, Azure Data Bricks** and **Azure Cosmos DB**.

TECHNICAL SKILLS

Cloud Platforms: AWS (S3, EC2, RDS, Redshift, Lambda), Azure, GCP (BigQuery, Dataflow, Pub/Sub)

Programming & Scripting

Languages: Python, SQL, Java, Scala, Shell scripting

Data Warehousing & ETL Tools:

Snowflake, Amazon Redshift, Google BigQuery, Azure Synapse, Apache Airflow, Talend, Informatica

Databases: MySQL, PostgreSQL, MongoDB, Cassandra, Oracle, SQL Server, DynamoDB

Big Data Technologies: Hadoop, Spark, Hive, HDFS, Kafka, Kinesis, Apache Beam

Data Visualization: Tableau, Power BI, Looker, Google Data Studio

DevOps & CI/CD: Jenkins, Docker, Kubernetes, Git, Terraform, Ansible, CI/CD pipelines

API & Automation: REST APIs, GraphQL, AWS Lambda, Azure Functions

Data Modeling: Star Schema, Snowflake Schema, Dimensional Modeling

Version Control: Git, GitHub, Bitbucket, GitLab

Data Security & Governance: Data encryption, GDPR/CCPA compliance, RBAC

Tableau, PowerBI, Informatica

EDUCATION:

- Masters from University of Cincinnati, USA.

WORK EXPERIENCE:

HP Inc, Dallas, Texas, USA | **Azure Data Engineer** | **Oct 2023 – Present**

OBJECTIVE: To utilize my expertise in data engineering, cloud technologies, and data pipeline development to help HP Inc. drive data-driven innovation and enhance business operations. By optimizing data workflows, ensuring data quality, and leveraging cutting-edge technologies, I aim to support HP's mission of delivering sustainable and impactful solutions to customers worldwide.

Key Responsibilities and Achievements:

- Integrated and stream inventory and sales data via **Kafka to Databricks** and apply aggregation or ML models to forecast **stock demand and availability** in near real-time.
- Ingested competitor pricing and real-time sales signals to build data pipelines that feed **machine learning pricing models** and update product prices dynamically.
- Optimizing data processing performance through **tuning** and monitoring involves identifying and **resolving bottlenecks** in the data pipeline and optimizing the data processing algorithms.
- Developed **Databricks ETL** pipelines using **notebooks, Spark Data frames, SPARK SQL and python scripting**.
- Develop and deploy **real-time data streams** to detect fraudulent patterns using **structured streaming with Delta Lake** and anomaly detection algorithms
- Design and manage structured pipelines across **Bronze (raw), Silver (cleaned), and Gold (aggregated)** Delta tables using **Databricks notebooks or workflows**.
- Built **PySpark** code to validate the data from raw source to **Snowflake tables**.
- Experience with **Azure** data storage solutions such as **Azure SQL Database, Azure Cosmos DB, and Azure Data Lake Storage** and Knowledge of data integration tools such as **Azure Data Factory and Azure Databricks**.
- Familiarity with data processing frameworks such as **Apache Spark and Hadoop** and Understanding of data modeling and schema design principles

Environment: Apache Spark, Hadoop, Azure SQL Database, Azure Cosmos DB, Azure Data Lake Storage, Spark Cluster, CI/CD, Jenkins, Snowflake tables, PySpark, python scripting, Linux.

Southwest Airlines, Dallas, Texas, USA | **Associate Data Engineer** | **Nov 2022 – Sept 2023**

OBJECTIVE: Southwest Airlines enhances its data infrastructure and drives actionable insights. By optimizing data pipelines, improving data quality, and implementing scalable solutions, I aim to support Southwest's mission of delivering exceptional customer experiences and efficient operational performance across its network.

Key Responsibilities and Achievements:

- Develop and implement data infrastructure, databases, and processing systems that are scalable, reliable, and efficient.
- Manage and optimize the flow of data from multiple sources and implement processes for **data modeling, mining, and production**.
- Implementing **ETL** Processes and Extract, Transform, Load (ETL) processes are crucial for moving data from various sources into a data warehouse and **AWS Data Engineers** use tools such as **AWS Glue** to automate these **ETL processes**.
- Perform data transformation and cleansing using **AWS Glue, AWS Lambda, or AWS EMR** and Implement real-time data streaming and processing using **Amazon Kinesis** and **AWS Lambda**.
- Responsible for creating **Hive tables**, loading data, and writing **hive queries** and **Configured Spark streaming** to get ongoing information from the **Kafka**.
- Good Knowledge on architecture and components of **Spark**, and efficient in working with **Spark Core, Spark SQL, Spark streaming** and expertise in building **PySpark** and **Spark-Scala** applications for interactive analysis, batch processing and stream processing.

- Provisioned high availability of **AWS EC2 instances**, migrated legacy systems to **AWS**, and developed **Terraform** plugins, modules, and templates for automating **AWS** infrastructure.
- Performed end-to-end Architecture & implementation assessment of various **AWS** services like **Amazon EMR, Redshift, S3**.
- Implemented the machine learning algorithms using python to predict the quantity a user might want to order for a specific item so we can automatically suggest using kinesis **firehose** and **S3** data lake and Used **AWS EMR** to transform and move large amounts of data into and out of other **AWS data stores** and **databases such as Amazon Simple Storage Service (Amazon S3) and Amazon DynamoDB**.
- Used Spark **SQL for Scala & amp, Python** interface that automatically converts **RDD** case classes to schema **RDD**.
- Import the data from different sources like **HDFS/HBase** into **Spark RDD** and perform computations using **PySpark** to generate the output response.

Environment: AWS EKS, GCP GKE, Kubernetes, Spark, Hadoop, Big Data, PySpark, Kafka, AWS Lambda, Amazon Kinesis, AWS Glue, AJAX, JSON, Java script, Hive tables, Tableau, Kibana, Kubernetes, Terraform

Telstra (Cognizant), Chennai, India | GCP Data Engineer | Oct 2021 – Aug 2022

OBJECTIVE: Telstra Corporation Limited is Australia's largest telecommunications and technology company, providing a wide range of services. I ensured accuracy and integrity of data and applications through analysis, coding, documenting, testing, and problem solving.

Key Responsibilities and Achievements:

- Migrating an entire oracle database to BigQuery and using of power bi for reporting and Build data pipelines in airflow in **GCP** for **ETL** related jobs using different airflow operators and Experience in **GCP Dataproc, GCS, Cloud functions, BigQuery**.
- Used **cloud shell SDK** in **GCP** to configure the services **Data Proc, Storage, BigQuery**
- Designed and Co-ordinated with **Data Science** team in implementing Advanced Analytical Models in **Hadoop** Cluster over large Datasets.
- Wrote scripts in **Hive SQL** for creating complex tables with high performance metrics like partitioning, clustering.
- Worked with **Google** data catalog and other **Google cloud API's** for monitoring, query and billing related analysis for **BigQuery**
- Hands of experience in **GCS bucket, G - cloud function, cloud dataflow, Pub/sub** cloud shell, **GSUTIL, BQ** command line utilities, **Data Proc, Stack driver**
- Coordinated with team and Developed framework to generate Daily adhoc reports and Extracts from enterprise data from **BigQuery**.
- Worked on creating **POC** for utilizing the **ML models** and **Cloud ML** for table Quality Analysis for the batch process. and Knowledge about cloud dataflow and **Apache beam**.

ENVIRONMENT: GCP Dataproc, GCS, BigQuery, Azure Data Factory, cloud shell, Data Science, Hadoop, Hive SQL, ETL, Google cloud API's, ML models, Apache beam, Pig, Hive, SQOOP

Divis Laboratories, Hyderabad, India | Data Engineer | Apr 2019 – Sept 2021

OBJECTIVE: Divis Laboratories is a leading Indian pharmaceutical company, specializing in the manufacture of active pharmaceutical ingredients. Orchestrated and optimized data pipelines to streamline the extraction, transformation, and loading (ETL) of data from diverse sources, utilizing tools and technologies aligned with the company's tech ecosystem.

Key Responsibilities and Achievements:

- Develop and manage **ETL** processes and automate operations using **SQL** Server Integration Services. Build dashboards in Tableau with **ODBC** connections from sources like **BigQuery/Presto SQL**, and create stored procedures in MS SQL for data retrieval and processing via **FTP**.
- Create Databricks **ETL** pipelines using notebooks, **Spark DataFrames, Spark SQL, and Python scripting**.

- Utilize **Tableau** for business intelligence and data visualization, connecting to various data sources to build comprehensive graphs and dashboards.
- Design and implement infrastructure as code using Terraform to automate provisioning and scaling of **cloud resources on Azure**.
- Work with **SAP SD** Module for customer management and sales reporting. Design and configure databases and backend applications, manage large datasets using **Pandas and SQL**, and maintain infrastructure as code with templates.
- Build and maintain **Docker** container clusters managed by **Kubernetes**. Implement **continuous delivery (CI/CD)** pipelines with Docker for custom application images using Jenkins.
- Perform data transformation and cleansing with **SQL queries, Python, and PySpark**. Use **HiveSQL, Presto SQL**, and **Spark SQL** for ETL jobs, applying the most suitable technology for efficient data processing.
- Process and load bound and unbound Data from Google pub/sub topic to **Bigquery** using **cloud Dataflow** with **Python**
- Implemented **Apache Airflow** for authoring, scheduling and monitoring **Data Pipelines**
- Proficient in Machine Learning techniques (Decision Trees, Linear/Logistic Regressors) and **Statistical Modeling**

Environment: ETL, SQL, ODBC, Big Query, Presto SQL, FTP, Spark Data Frames, Spark SQL, and Python scripting, Tableau, Azure, SAP SD, Pandas, Docker, Kubernetes, CI/CD, HiveSQL, Presto SQL, Pub/sub, Apache Airflow